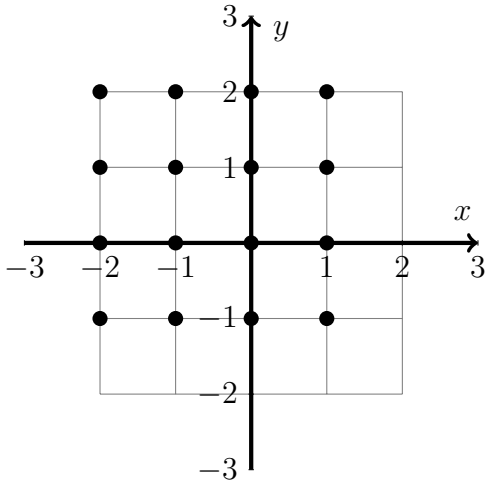
...

Double Integrals

Consider $f(x, y) = x^2 + y^2$. The graph is a paraboloid with a vertex at the origin. Suppose we'd like to know how much volume is bounded between the paraboloid and the xy-plane over the square $S = [-2, 2] \times [-2, 2]$.

Let's estimate it.

Here is one particular estimate: We can split S into 16 sub-rectangles, each of area $\Delta A = 1$, using gridlines at $x = -1, 0, 1$ and at $y = -1, 0, 1$, and then evaluate f at upper-left corners ("sample points") to get an estimate of the value of f over each sub-rectangle.



So an estimate for the volume is given by evaluating f at each sample point and multiplying that value of the area of the sub-rectangle:

$$V \approx (8 + 5 + 4 + 5 + 5 + 2 + 1 + 2 + 4 + 1 + 0 + 1 + 5 + 2 + 1 + 2) \cdot 1 = 48.$$

Such a sum is called a Riemann sum.

Let $f(x, y)$ be defined over a rectangular region R in the xy-plane. If the limit,

$$\lim_{n\to\infty, \Delta A_i \to 0} \sum_{i=1}^{n} f(x_i, y_i)\Delta A_i,$$

exists for all partitions of R into n sub-rectangles, where (x_i, y_i) is a sample point from the ith sub-rectangle, and ΔA_i (such that $\Delta A_i \to 0$) represents the area of the ith sub-rectangle, then the limit is the double integral of f over R.

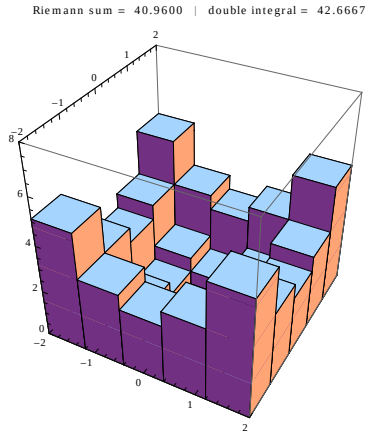
Notation: $\int\int_R f(x, y)\, dA$.

So on the previous page, 48 is an estimate of $\int\int_{[-2,2]\times[-2,2]} x^2 + y^2\, dA$.

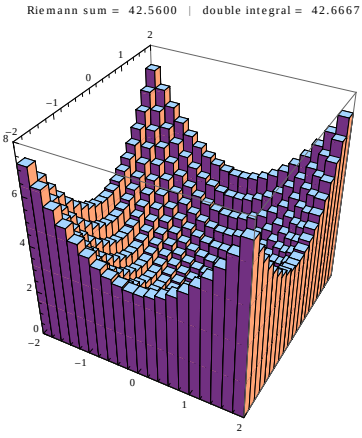
Since f may sometimes be negative on R, the double integral is the net volume of f over R, which is the volume between $z = f(x, y)$ and $z = 0$ when f is non-negative MINUS the volume between $z = f(x, y)$ and $z = 0$ when f is negative.

Of course the Riemann sum estimates for double-integrals get better with more rectangles! In the following figures, we visualize Riemann sum estimates for $\int\int_{[-2,2]\times[-2,2]} x^2 + y^2\, dA$ made with various numbers of equal-sized sub-rectangles, and rectangle centers as sample points.

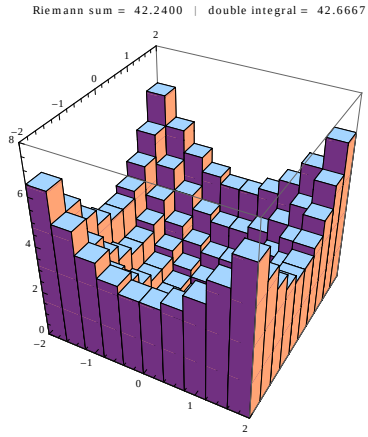
25 sub-rectangles:



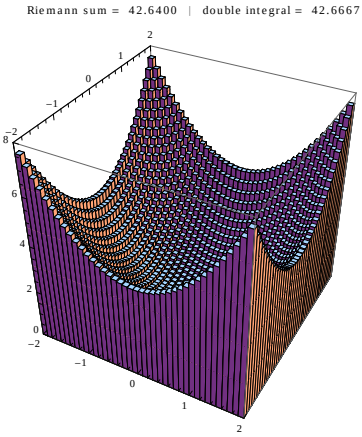
400 sub-rectangles:



100 sub-rectangles:



1600 sub-rectangles:



Fubini's Theorem (over rect. regions): Suppose $f(x,y)$ is continuous over a rectangular region $R = [x_1, x_2] \times [y_1, y_2]$. Then $\int\int_R f(x,y)\,dA = \int_{x_1}^{x_2}\int_{y_1}^{y_2} f(x,y)\,dydx = \int_{y_1}^{y_2}\int_{x_1}^{x_2} f(x,y)\,dxdy$.

(The proof is skipped.)

Sometimes we refer to this process as "iterating the integral" and refer to integrals inside of integrals as "iterated integrals."

For example,

$$\int\int_{[-2,2]\times[-2,2]} x^2 + y^2\,dA = \int_{-2}^{2}\int_{-2}^{2} x^2 + y^2\,dydx = \int_{-2}^{2}\left(x^2 y + \frac{1}{3}y^3\Big|_{y=-2}^{y=2}\right)dx =$$

$$= \int_{-2}^{2}\left(4x^2 + \frac{16}{3}\right)dx = \frac{4}{3}x^3 + \frac{16}{3}x\Big|_{-2}^{2} = \frac{128}{3} \approx 42.6667.$$

Let's evaluate $\iint_{[0,1]\times[1,2]} \frac{x}{(1+xy)^2}\,dA$.

First we choose an order for the iteration. Since it appears to be easier to integrate with respect to y first, we choose the $dydx$ order:

$$\int_0^1 \int_1^2 \frac{x}{(1+xy)^2}\,dydx = \int_0^1 -\frac{1}{1+xy}\Big|_{y=1}^{y=2} dx = \int_0^1 \frac{1}{1+x} - \frac{1}{1+2x}\,dx = \ln(2) - \frac{1}{2}\ln(3) = \ln\left(\frac{2}{\sqrt{3}}\right).$$

The definition of a double integral for a rectangular region can be extended to general regions that are closed and bounded, and a generalized version of Fubini's Theorem can be applied. There are two basic types of regions:

(1) R is bounded by $a \le x \le b$ and $g(x) \le y \le h(x)$. Then if $f(x,y)$ is continuous over R,
$$\iint_R f(x,y)\,dA = \int_a^b \int_{g(x)}^{h(x)} f(x,y)\,dydx.$$

(2) R is bounded by $c \le y \le d$ and $\ell(y) \le x \le m(y)$. Then if $f(x,y)$ is continuous over R,
$$\iint_R f(x,y)\,dA = \int_c^d \int_{\ell(y)}^{m(y)} f(x,y)\,dxdy.$$

Notes:

-A region might need to be split into more than one of these types of regions above! If a region R is partitioned into subregions R_1 and R_2 then
$\iint_R f(x,y)\,dA = \iint_{R_1} f(x,y)\,dA + \iint_{R_2} f(x,y)\,dA.$

-If the integrand is 1, the double integral recovers the area of the region (since it's the volume under a surface that is height 1 over the region).

-The average value of $f(x,y)$ over a region R is $\frac{1}{A}\iint_R f(x,y)\,dA$ where A is the area of R.

Let's evaluate $\int_0^4 \int_{x/2}^2 e^{-y^2}\,dydx$.

This one is already presented as an iterated integral, but since we cannot find an elementary antiderivative of e^{-y^2} with respect to y we switch the order of integration!

The region described by $0 \le x \le 4$ and $\frac{x}{2} \le y \le 2$ can also be described by $0 \le y \le 2$ and $0 \le x \le 2y$.

So $\int_0^4 \int_{x/2}^2 e^{-y^2}\,dydx = \int_0^2 \int_0^{2y} e^{-y^2}\,dxdy = \int_0^2 2ye^{-y^2}\,dy = -e^{-y^2}\Big|_0^2 = 1 - \frac{1}{e^4}.$